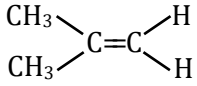
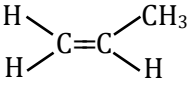
RACE
Regular Analysis Through Continuous Exercise

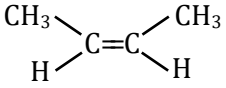
Geometrical Isomerism and Conformations : RACE-01

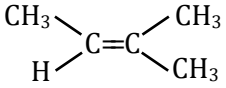
Geometrical Isomerism in different systems

1. Which can show 'Geometrical isomerism' ?

(A) 

(B) 

(C) 

(D) 

CGI074

2. 2–butene shows geometrical isomerism due to :–

(A) Restricted rotation about double bond (B) Free rotation about double bond

(C) Free rotation about single bond (D) Chiral carbon

CGI075

3. Which compound does not give geometrical isomerism:-

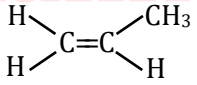
(A) Pent–2–ene (B) 1–chloro propene

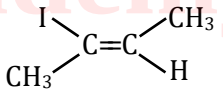
(C) 3–chloro propene (D) 1,2–dibromo ethylene

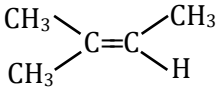
CGI076

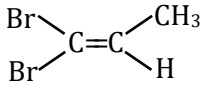
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4. Geometrical isomerism is shown by :–

(A) 

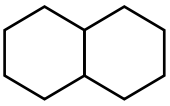
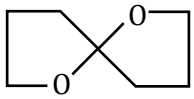
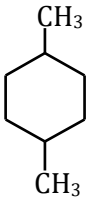
(B) 

(C) 

(D) 

CGI077

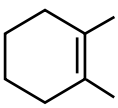
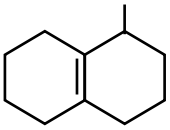
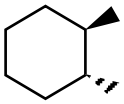
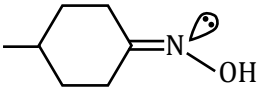
5. Which of the following compound will not shows geometrical isomerism ?

(A) Ph –CH =N –Me (B)  (C)  (D) 

CGI084

Nomenclature of Geometrical Isomers

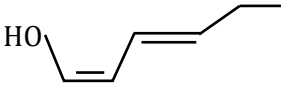
6. Which of the following cannot be represent in E or Z

(A)  (B)  (C)  (D) 

CGI087

For More Join: @IITJEE_Advanced

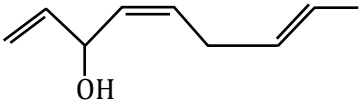
7. The correct stereochemical descriptions for the structure given below are :



(A) 1 Z, 3Z (B) 1Z, 3E (C) 1E, 3E (D) 1E, 3Z

CGI091

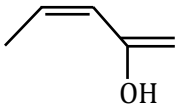
8. Which of following is correct for given structure



(A) 4Z , 7E (B) 4E, 7Z (C) 2E, 5Z (D) 2Z, 5E

CGI092

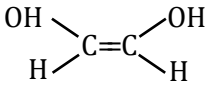
9. Which of the following is correct for given structure.



(A) 3Z (B) 3E (C) 1Z,3Z (D) 1Z, 3E

CGI093

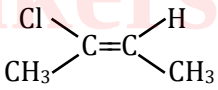
10. Assign cis/trans isomerism to the following



(A) cis (B) trans (C) E (D) none of these

CGI099

11. Assign cis/trans isomerism to the following



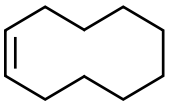
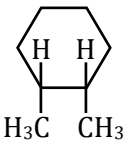
(A) cis (B) trans (C) Z (D) none of these

CGI100

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Geometrical Isomerism in different systems and others

12. Which will show geometrical isomerism :-

(A) C_6H_5–CH=NOH (B)  (C) C_6H_5 –N=N–C_6H_5 (D) 

CGI085

13. Which of the following statements is/are not correct :

(A) Metamerism belongs to the category of structural isomerism

(B) Tautomeric structures are the resonating structures of a molecule

(C) Keto form is always more stable than the enol form

(D) Geometrical isomerism is shown only by alkenes

CGI101

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	C	B	C	D	B	A	A	A
Question	11	12	13							
Answer	A	ABCD	BCD							

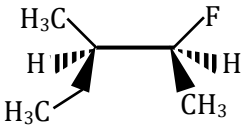
For More Join: @IITJEE_Advanced

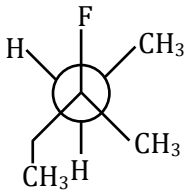
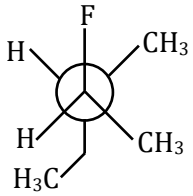
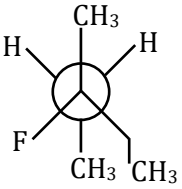
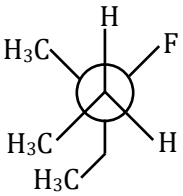
RACE
Regular Analysis Through Continuous Exercise

Geometrical Isomerism and Conformations : RACE-02

Conformers of different systems and their stability

1. Which is the most stable conformer along the 2,3 C–C bond axis of the compound ?



(A) 
(B) 
(C) 
(D) 

CGI110

2. Most stable conformation of butane is :-

(A) Partial eclipsed (B) Fully eclipsed (C) Staggered (D) Gauche

CGI112

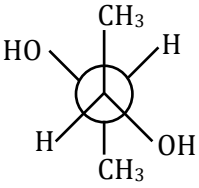
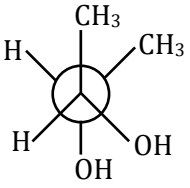
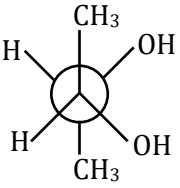
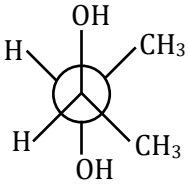
3. What is dihedral angle in staggered form of ethane :-

(A) 30° (B) 45° (C) 75° (D) 60°

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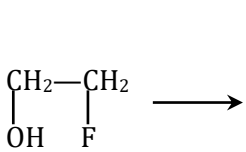
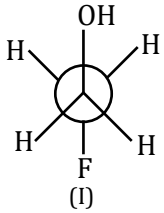
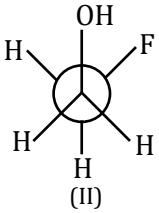
CGI113

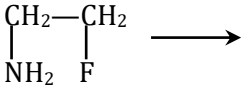
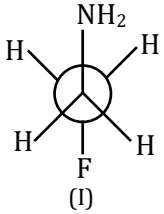
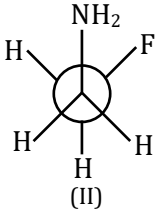
4. Which one of the following is the most stable conformer?

(A) 
(B) 
(C) 
(D) 

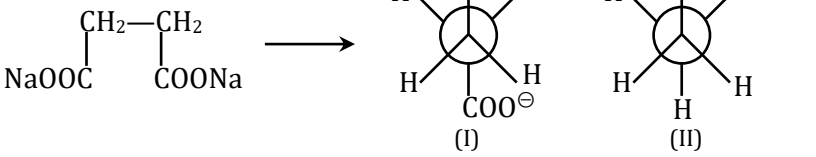
CGI117

5. Which of the following 1st is most stable form then 2nd form?

(A) 

(I)

(II)

(B) 

(I)

(II)

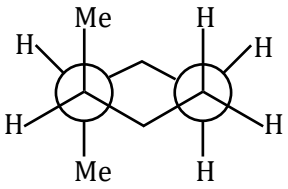
For More Join: @IITJEE_Advanced

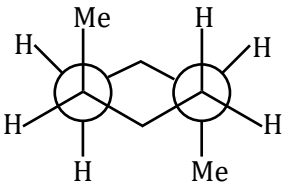
(C) 

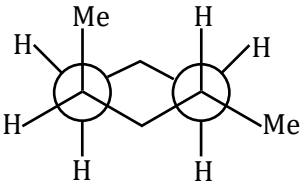
(D) None of these

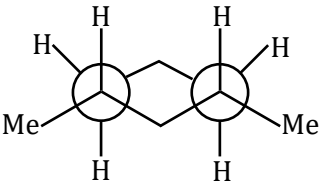
CGI119

6. Which of the following isomeric structure have lowest energy ?

(A) 

(B) 

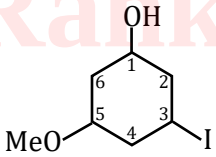
(C) 

(D) 

CGI122

7. Which is the most stable conformation of

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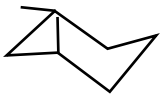
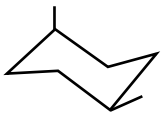
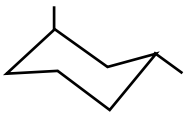


(Where –OH, –I, – OMe at C–1, C–3 and C–5 positions)

(A) e, e, a (B) a, e, a (C) e, e, e (D) a, e, e

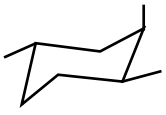
CGI125

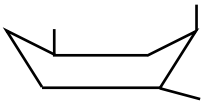
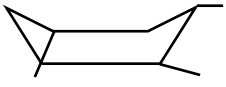
8. Which of the following compounds is most stable ?

(A)  (B)  (C)  (D) 

CGI127

9. Among the structure shown below, which has lowest potential energy ?

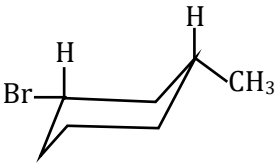
(A)  (B) 

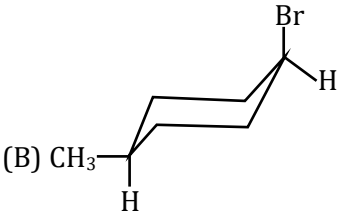
(C)  (D) 

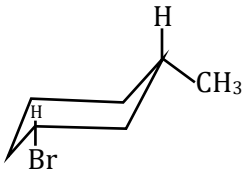
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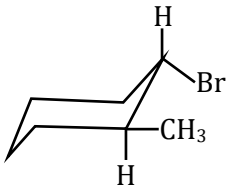
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10. Determine cis isomers among following.

(A) 

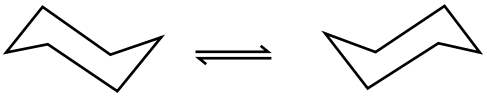
(B) 

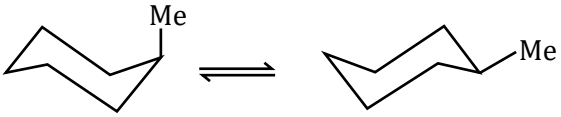
(C) 

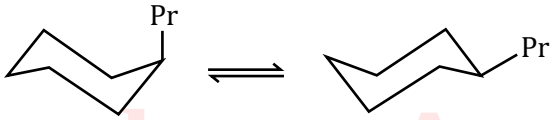
(D) 

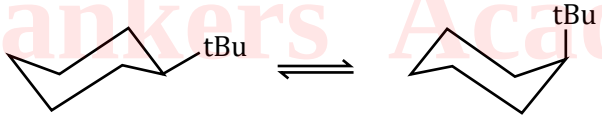
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11. Among the following equilibrium constant is greater than 1 for.

(A) 

(B) 

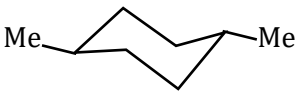
(C) 

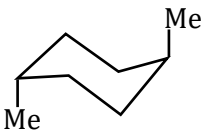
(D) 

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12. Select the correct statement for the following species.

(I) 

(II) 

(III) 

(IV) 

(A) I & II are geometrical isomers
(B) II & III are geometrical isomers
(C) III & IV are conformers
(D) I & IV are conformers

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	D	C	C	D	B	D	A	ABC
Question	11	12								
Answer	BC	BC								

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